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NYC Trip Duration Project

- Predicts trip durations in New York using 10 features and 1M training samples.
- Best model achieved with **XGBoost Regressor** — F1-score: 76% (Val) & 72% (Test).
- Frontend built with **Streamlit** and backend powered by **FastAPI**.

Credit Card Fraud Detection

- Classifies fraudulent transactions using 31 features and 56,960 examples (with 28 anonymized).
- Handled high class imbalance (0.172% positive ratio) focusing on **F1-score** and recall.
- Achieved top performance with **Random Forest**: Val_F1 = 89%, Test_F1 = 80%.

Data analysis

Superstore EDA

- In-depth EDA of the Superstore Dataset using a **Jupyter Notebook**, covering sales, profit, categories, regions, and customer segments.
- Uncovered performance trends and feature relationships to drive actionable business insights.
- Visualizations highlight patterns across time, geography, and product lines.

Rag

Mini_Rag Application

- Built a minimal, educational RAG (Retrieval-Augmented Generation) question answering system.
- Integrated LLMs, vector databases, and backend tools like **FastAPI**, **Docker**, **Celery**, and **PostgreSQL**.

- Step-by-step guidance included for learning purposes.

Automation (n8n)

Job Application Tracker

- Automated job application tracking system built with **n8n**.
- Monitors Gmail for sent applications and incoming replies, logs everything to **Google Sheets**.
- Sends instant **Telegram** notifications — with zero manual data entry.

AI Newsletter Digest

- Pulls articles daily from 5–10 **RSS feeds** (tech, news, niche topics) via n8n.
- Sends each article to an AI to produce a 2-sentence summary, then groups them by topic.
- Delivers a structured, noise-free digest straight to your **email** every morning.

Social Media Content Pipeline

- Automated content pipeline built with **n8n** transforming raw ideas from **Obsidian** into platform-specific posts.
- Generates optimized content for **LinkedIn**, **Facebook**, and **X (Twitter)**.
- Includes a human-in-the-loop review step before publishing to ensure quality and control.

Customer Support Auto-Responder

- AI-powered **n8n** workflow that classifies incoming support emails automatically.
- Instantly replies to FAQs in under a minute while escalating complex cases to human agents.
- Maintains full context and logs for efficient support handling and team collaboration.
- **Overview:** Reduces manual triaging for small teams by automating repetitive support tasks while preserving human oversight.

Console projects (c++ & python)

Ask Me App

- Interactive **Python** console app where users ask questions and receive structured responses.
- Follows clean project architecture, virtual environment isolation, and Python best practices.
- Modular codebase designed for readability and easy extension.

Contact Management System

- **C++** system to store, update, delete, and manage contact details efficiently.
- Leverages a custom **templated Array** data structure for dynamic storage.
- Advanced features include reverse sorting and persistent save/load to file.

Employee Manager

- **Python** console project with a clean separation of concerns across EmployeeManager.py and FrontEndManager.py.
- Backend handles all business logic for managing employee records.
- Frontend layer provides a user-friendly console interface.

Expedia Project

- **Python** travel-management console app structured with dedicated backend, frontend, and driver.py entry point.
- Simulates booking and itinerary management workflows.
- Clear layered architecture separating data logic from user interaction.

Library Management System

- **Python** library app split across BackEndLibrary.py, FrontEndLibrary.py, and LibraryManager.py.
- Handles book cataloguing, borrowing, and returning operations.
- Includes a dedicated test file (LibraryManager_Tests.py) for unit validation.

Payroll Project

- **Python** payroll system covering employee records, invoice generation, and payroll printing.
- Modular design across `Employees.py`, `Invoices_Payroll.py`, `Payroll_Printing.py`, and `Validations.py`.
- Input validation layer ensures data integrity throughout the workflow.

Hospital Management System

- **Python** hospital management app structured into backend, frontend, models, and tests layers with a `main.py` entry point.
- Manages patient records, appointments, and hospital operations with full test coverage.
- Documented with detailed project spec PDFs included in the repository.

Hackathons

Will It Rain On My Parade?

- Backend for a weather likelihood app reporting chances of adverse conditions (heat, cold, rain, wind, humidity) for a user-selected location and date.
- Built with **Laravel 11** (PHP 8.2), **PostgreSQL**, and the **Open-Meteo** forecast & geocoding APIs; supports city name or lat/lon input up to +14 days ahead.
- Personal role: weather data analysis and insights extraction.